

Arjuna JEE 2025

Mathematics

Binomial Theorem

DPP: 4

Q1 The sum of the coefficients of the polynomial

$$(1 + x - 3x^2)^{2163} \text{ is}$$

- (A) 1 (B) -1
(C) 0 (D) 2

Q2 If $n \in \mathbb{N}$ and n is even

$$\frac{1}{1 \cdot (n-1)!} + \frac{1}{3! \cdot (n-3)!} + \dots + \frac{1}{(n-1)! \cdot 1!}$$

- (A) 2^n
(B) $\frac{2^{n-1}}{n!}$
(C) $2^n \cdot n!$
(D) None

Q3 The sum of binomial coefficient of $(2x + \frac{1}{x})^n$ is equal to 256. The constant term in the expansion is

- (A) 1120 (B) 2110
(C) 1210 (D) None

Q4 ${}^{15}C_3 + {}^{15}C_5 + \dots + {}^{15}C_{15} =$

- (A) 2^{14}
(B) $2^{14} - 15$
(C) $2^{14} + 15$
(D) $2^{14} - 1$

Q5 $C_1 + 2C_2 + 3C_3 + 4C_4 + \dots + nC_n =$

- (A) 2^n
(B) $n \cdot 2^n$
(C) $n \cdot 2^{n-1}$
(D) $n \cdot 2^{n+1}$

Q6 Find the value of

$$1 \cdot 2C_2 + 2 \cdot 3C_3 + 3 \cdot 4C_4 + \dots + n(n-1)C_n$$

Q7 The sum

$$S = {}^{20}C_2 + 2 \cdot {}^{20}C_3 + 3 \cdot {}^{20}C_4 + \dots + 19 \cdot {}^{20}C_{20}$$

- (A) $1 + 5 \cdot 2^{20}$
(B) $1 + 2^{21}$
(C) $1 + 9 \cdot 2^{20}$
(D) 2^{20}

Q8 If $C_0, C_1, \dots, C_n$ denote the binomial coefficients in the expansion of $(1+x)^n$. Then, the value of $C_1 - 2C_2 + 3C_3 - 4C_4 + \dots$ (upto n terms) is

- (A) 2^n (B) 2^{-n}
(C) 0 (D) 1

Q9 If $C_0, C_1, C_2, \dots, C_n$ denote the binomial coefficients in the expansion of $(1+x)^n$, then $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1}$ is equal to

- (A) $\frac{2^{n+1}-1}{n+1}$
(B) $\frac{2^n-1}{n}$
(C) $\frac{2^{n-1}-1}{n-1}$
(D) $\frac{2^{n+1}-1}{n+2}$

Q10 If the sum of the coefficients in the expansion of $(\alpha^2 x^2 - 2\alpha x + 1)^{51}$ vanishes, then the value of α , is

- (A) 2 (B) -1
(C) 1 (D) -2

Q11 The sum of all coefficients in $(2x^2 - 3x + 1)^{11}$, is

- (A) 0 (B) -1
(C) 1 (D) 2



- Q12** Maximum sum of the coefficients in the expansion of $(1 - x \sin \theta + x^2)^n$ is
 (A) 1 (B) 2^n
 (C) 3^n (D) 0
- Q13** The sum of the coefficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$, is
 (A) 2.6^{10}
 (B) 3.6^{10}
 (C) 6^{11}
 (D) None of these
- Q14** The last two digits of 3^{400} are
 (A) 01 (B) 29
 (C) 43 (D) 81
- Q15** The last two digits of 21^{100} , is
 (A) 00 (B) 01
 (C) 11 (D) 21
- Q16** Find the last three digits in 11^{50} .
- Q17** If $C_r = {}^{25}C_r$ and $C_0 + 5 \cdot C_1 + 9 \cdot C_2 + \dots + (101) \cdot C_{25} = 2^{25} \cdot k$, then k is equal to
- Q18** If I is integral part of $(2 + \sqrt{3})^n$ and f is its fractional part. Then $(I + f)(1 - f)$ is
 (A) $I + 1$ (B) 1
 (C) n (D) 2^n
- Q19** The sum of the series $\sum_{r=0}^{10} {}^{20}C_r$, is
 (A) 2^{20}
 (B) 2^{19}
 (C) $2^{19} + \frac{1}{2} {}^{20}C_{10}$
 (D) $2^{19} - \frac{1}{2} {}^{20}C_{10}$
- Q20** ${}^nC_0 - \frac{1}{2} {}^nC_1 + \frac{1}{3} {}^nC_2 - \dots + (-1)^n \frac{{}^nC_n}{n+1}$
 $\stackrel{=}{=} \text{(A) } n$

- (B) $1/n$
 (C) $\frac{1}{n+1}$
 (D) $\frac{1}{n-1}$



Answer Key

Q1 (B)

Q2 (B)

Q3 (A)

Q4 (B)

Q5 (C)

Q6 $n(n-1).2^{n-2}$

Q7 (C)

Q8 (C)

Q9 (A)

Q10 (C)

Q11 (A)

Q12 (C)

Q13 (B)

Q14 (A)

Q15 (B)

Q16 001

Q17 51

Q18 (B)

Q19 (C)

Q20 (C)

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